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27 September 1985
SUPERSEDING
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27 June 1968

MILITARY SPECIFICATION

POSITIONERS, WELDING, ROTATING TABLE TYPE, CANTILEVER ARM DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers table type welding positioners designed with the table off-set from the base to provide 360 degree table rotation. This specification covers both manual and powered table rotation, and both bench mounted and floor mounted positioners.

1.2 Classification. This specification covers positioners of the following classes and sizes. The class and size to be furnished shall be as specified (see 6.2.1).

Class I - Manual table rotation with manual tilt.

Class II - Powered table rotation with manual tilt.

Size 1	-	50 pound (22 Kg)	capacity, bench mounting.
Size 1a	-	100 pound (45 Kg)	capacity, bench mounting.
Size 2	-	200 pound (91 Kg)	capacity, bench mounting.
Size 3	-	300 pound (136 Kg)	capacity, roller base mounting.
Size 4	-	500 pound (227 Kg)	capacity, floor mounting.
Size 5	-	1000 pound (454 Kg)	capacity, floor mounting.
Size 6	-	2000 pound (907 Kg)	capacity, floor mounting.
Size 7	-	4000 pound (1814 Kg)	capacity, floor mounting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3436

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Class III - Powered table rotation with powered tilt.

Size 5 - 1000 pound (454 Kg) capacity, floor mounting.
Size 6 - 2000 pound (907 Kg) capacity, floor mounting.
Size 7 - 4000 pound (1814 Kg) capacity, floor mounting.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NEMA ICS 1 - Industrial Control and Systems.

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

(Applications for copies should be addressed to the Sales Dept., American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected for first article inspection (see 4.3 and 6.3).

3.2 Design. The positioner shall be new and one of the manufacturer's current models capable of performing welding positioner operations in accordance with the requirements herein. The positioner shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified herein that are a part of the manufacturer's current model shall be included in the positioner furnished.

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The positioner shall be of the cantilever arm design, with the arm supported at one end only from the upper end of the vertical column. The work table shall be located on the projecting end of the arm off-set 90 degrees from the column. Each positioner work table shall provide tilt action of 360 degrees around the arm axis and 360 degrees table rotation at any tilt position. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Bench mounted positioners. Sizes 1, 1a, and 2 positioners shall be configured for bench mounting. Tilt action around the arm axis shall be accomplished through the arm mounting support. With the work table positioned face up and at 0 degrees tilt, the horizontal plane of the table top shall be higher than any other portion of the positioner. The base and column assembly shall provide for the arm to revolve 360 degrees around the base with means for locking in any position.

3.2.2 Roller supported positioner. The size 3 positioner shall be mounted on a "T" type base. The cross section of the base shall be approximately 35 inches in length and the supporting section approximately 30 inches long by 10 inches wide. The positioner column shall be mounted at the center of the cross section of the "T". The base shall be mounted on 3 inch diameter steel casters with 2 swivel casters located on the cross section and 2 rigid casters located on the supporting section. When specified (see 6.2.1), foot operated locks shall be provided for each swivel caster. Tilt action, horizontal plane table top height, and arm rotation requirements for the size 3 positioner shall be the same as specified for sizes 1 through 2 positioners.

3.2.3 Floor mounted positioners. The sizes 4 through 7 positioners shall be configured for floor mounting. The arm shall be hinged at two points to provide compensating angle adjustments of from 0 degrees to not less than 30 degrees. One hinge point shall be through the arm mounting support and one hinge point shall be between the arm mounting support and work table. Table tilt action shall be accomplished through the portion of the arm mounted to the column. The hinge arrangements shall be such that the work table can be adjusted to the horizontal position regardless of the degree of angle. The arm shall revolve 360 degrees around the base and shall have means for locking in any position.

3.2.4 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the positioner. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

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3.2.5 Reclaimed materials. The positioner may contain reclaimed materials provided such materials will not jeopardize the positioner's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the positioner.

3.2.6 Energy efficiency. The positioner and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.7 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.8 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the positioner that present safety hazards. The safety devices shall not interfere with the operation of the positioner. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All positioner parts, components, mechanisms, and assemblies furnished on the positioner whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the positioner itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.9 Mercury restriction. The positioner shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.10 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the positioner.

3.2.11 Environmental protection. The positioner shall be so designed and constructed that, under the operating, service, transportation and storage conditions described herein, the positioner shall not emit materials hazardous to the ecological system as prescribed by federal, state, or local statutes in effect at the point of installation.

3.2.12 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.13 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or positioner.

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3.3 Construction. The positioner shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the positioner to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the positioner. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the positioner shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter and at least one coat of primer and one coat of finish color is applied.

3.3.6 Threads. All threaded parts used on the positioner and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.4 Components. The basic components of the positioner shall consist of at least a base and column, an arm, a circular flat worktable, and a variable speed motorized mechanism for the powered rotating table positioners.

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3.4.1 Frame and base assembly. The frame and base assembly shall be a rigid steel weldment constructed to maintain original alignment and to minimize deflection. The frame shall consist of a base, column, and arm assembly. Floor mounted positioners shall be capable of supporting maximum loads at any tilt or angle position without the use of counterweights or floor attachments. The base of the positioner shall be provided with means for bolting it to the floor or work bench. Size 1, 1a, 2, and 7 positioners shall have either a fixed or adjustable column. Size 3, 4, 5, and 6 positioners shall provide for vertically adjusting the table height. The vertical adjustment shall be in addition to the arm angle adjustment. The table height shall be measured with the angle adjustment at "0" degrees and with the arm and table on a level plane. Each positioner shall be provided with an electrical ground block on the base or column suitable for bolting an electrical ground cable.

3.4.2 Positioner table. All positioner tables shall be circular with a flat work surface machined to within 3/32 of an inch in flatness of the surface. The table shall be supported on a spindle. The spindle shall be supported by sealed anti-friction bearings. The class I positioners shall be provided with a hand operated adjustable friction brake system and an indexing pin locking device for locking the table in at least eight different equally spaced positions. The table for class II and class III positioners shall be provided with a dynamic braking system and shall be powered by an electric motor. Each table shall be provided with either four radial T slots conforming to ANSI B5.1 or four equally spaced unthreaded bolt holes for mounting the workload.

3.4.3 Motorized table rotation. The table motorized drive system for classes II and III positioners shall be powered by a variable speed, silicon control rectifier (SCR), direct current (DC) motor of the totally enclosed type. The motor shall provide dynamic braking with overload protection. The table rotation shall be reversible and the speed control shall be accurate within 3 percent of the base speed. The drive system shall not restrict the tilt action, angle adjustment, or reduce the capacity of the positioner. The self locking drive reducer shall be totally enclosed.

3.4.3.1 Controlled atmosphere operation table drive. The drive mechanism, controls, and lubrication system of the class II, size 1 positioners shall be completely enclosed and capable of operating in a controlled atmosphere chamber. Rotation accuracy shall be within 3 percent of the base speed. When specified (see 6.2.1), remote controls shall be furnished.

3.4.3.2 Normal operating table drive. The powered table rotation RPM range for class II and class III shall conform to table I. The positioner drive system shall be controlled by a remote foot operated control switch with forward, reverse, and stop operations. The remote foot operated switch shall be provided with not less than 8 feet of cable wired through an "on" and "off" switch located on the positioner. The forward and reverse selector switch for the powered table rotation may be located on the positioner or in the electrical cabinet.

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3.4.4 Tilt mechanism. The tilt portion of the arm shall be supported by sealed, anti-friction, tapered roller bearings. The tilt arm of each positioner, shall be provided with an indexing pin lock system for locking the tilt mechanism in at least eight different equally spaced positions. In addition to the pin lock system, the tilt arm of each positioner shall be provided with a hand operated, adjustable, friction brake system. The powered tilt mechanism shall be controlled by a forward and reverse toggle switch with a light to indicate when the power is on.

3.4.5 Angle adjustment system. The arm angle adjustments for the sizes 1a through 7 positioners shall be accomplished through the use of a self-locking worm and worm gear sector system. The system shall be manually adjustable and capable of being adjusted to any design angle with a capacity load under all operating conditions.

3.4.6 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Sizes 1 through 6 positioners shall operate from a single phase, 115 volt, 60 Hz circuit. The size 7 positioner shall operate from a 230/460 volt, 3 phase, 60 Hz circuit. The size 7 positioner shall be initially wired for operation on 230 volts. Each positioner shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the positioner shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79. A power cable of not less than 8 feet in length shall be furnished.

3.4.6.1 Weld current conduction and grounding device. All bearings and positioner adjustments shall be capable of withstanding ground weld current conduction amperes of not less than that specified in table I. Each positioner shall be equipped with a suitable ground cable attachment. The ground attachment shall be permanently mounted in a position to prevent interference throughout the full operational range of the positioner. The connection shall assure low resistance conduction.

3.4.6.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements of the totally enclosed type. Motors shall conform to ANSI/NEMA MG1, and all electronic controls and enclosures shall conform to ANSI/NEMA ICS 1.

3.4.6.3 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.6.4 Control circuit voltage. Auxiliary control circuits shall be isolated from the input power supply by a low-voltage transformer having a secondary voltage not greater than 115 volts.

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TABLE I. Positioner sizes and capacities.

Class I, II, and III positioners							Class II and III positioners		Class III positioners
Positioner size	Capacity load, pounds	Center of gravity (cg), inches	Table off-set, center of column to center of table, inches	Table diameter, inches	Table height, inches	Ground weld current conduction, amperes	Table rotation speed range, rpm	Table torque, inch-pounds	Table tilt torque, inch-pounds
1	50	4	10	5	10	300	0.2 to 10	50	---
1a	100	12	12	6	10	500	0.2 to 6.0	400	---
2	200	12	20	6	12	500	0.2 to 6.0	1500	---
3	300	12	20	6	22 to 31	500	0.2 to 6.0	1500	---
4	500	12	32	6	22 to 31	800	0.2 to 6.0	1500	---
5	1000	12	32	8	30 to 40	800	0.2 to 6.0	1500	6,000
6	2000	12	35	8	33 to 39	1200	0.2 to 6.0	1500	12,000
7	4000	12	51	12	41	2000	.031 to 1.23	2000	26,000

NOTE: All requirements shall be not less than the figures shown. Where a range is shown, the requirements shall range from not more than the minimum to not less than the maximum figures shown in the table

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3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the sizes and capacities of the positioners shall be not less than those specified in table I for the positioner class and size specified.

3.6 Performance. Each positioner shall be capable of rotating and tilting its capacity loads, and shall conform to the requirements of table I and all additional requirements of this specification. In addition, sizes 4 through 7 positioners shall be capable of balancing a weldment when the center of gravity of the weldment has been located directly on the rotation axis of the work table by providing necessary table tilt and arm angle adjustments so that the rotation axis will intersect the center of gravity of the weldment.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, one set of hand tools, normally furnished by the manufacturer, shall be furnished. Additional equipment required shall be furnished as specified and fully described (see 6.2.1).

3.8 Optional equipment. When specified (see 6.2.1), the following optional equipment shall be furnished and shall be fully described.

a. Air or hydraulic tilt and table brake with foot control valve.
b. Extra work tables.
c. Three jaw self-centering chuck.

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each positioner. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

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3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each positioner. The nameplate shall contain the information listed below. If the positioner is a special model, the model designation shall include the model of the basic standard positioner and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Date of manufacture
Table RPM (when applicable)
Contract Number or Order Number
National Stock Number or Plant Equipment Code

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract and shall be printed in the English language.

3.12 Workmanship. Workmanship of the positioner and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of

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the item to pass the examination and all tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) quality conformance inspection shall consist of at least the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.5 Examination. The positioner and its equipment shall be visually examined to determine compliance with all requirements of this specification.

4.6 Tests.

4.6.1 Operational test. Each positioner shall be operated through all movements without a table load. Table rotation, tilt action, and angle adjustments shall be smooth and shall not show evidence of any galling or binding. Tilt and table brake systems and pin locking devices shall be checked for operation. Motorized table positioners shall be connected to the specified source of electricity. The table speed, control range accuracy, and torque shall be checked for conformance to the requirements specified herein.

4.6.2 Performance tests. A capacity load, in accordance with table I, shall be secured to the table surface. All tests cited in 4.6.1 shall be repeated. With the table positioned over the base, floor mounted positioners shall be adjusted to the maximum vertical height and without the use of counterweights, shall be operated through all tilt and angle movements without tilting at the base. DC amperage equal to at least the ground weld current amperes specified for the size positioner shall be passed through the positioner with the table rotating for a period of at least one minute with no current interruption or damage to the positioner.

4.7 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The size 1 positioner is intended primarily for use in controlled atmosphere chambers to position and rotate workpiece weldments for extremely precise welding processes. Sizes 1a through 7 positioners are

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intended for use in production welding processes to position, balance, and rotate workpiece weldments for down-hand operator welding.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. Caster locks, if required (see 3.2.2).

e. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.4).

f. Additional safety and health requirements (see 3.2.8).

g. Painting, if different (see 3.3.5).

h. Remote control for size 1 positioner, if required (see 3.4.3.1).

i. Electrical system, if different (see 3.4.6).

j. Motor enclosure, if different (see 3.4.6.2).

k. Size and capacity, if different (see 3.5).

l. Additional performance features, if required (see 3.6.1).

m. Standard equipment, if different (see 3.7).

n. Optional equipment, when required (see 3.8).

o. Specify applicable repair parts, if required (see 3.9).

p. Lubrication chart or plate, if different (see 3.10.1).

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q. Nameplate, if different (see 3.10.2).

r. First article inspection, if different (see 4.3).

s. Quality conformance inspection, if different (see 4.4).

t. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, examination and test reports, and other instructions for operation and maintenance should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.8 requires compliance only with those OSHA requirements that concern the positioner itself. It does not require compliance with those OSHA requirements that concern "the positioner in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual positioners in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional positioner requirements that will integrate the new positioner into its future operating environment. The positioner shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial positioner supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (f).

6.5 Cross-reference of classification changes. Positioner classification changes from the previous issue of this specification are as follows:

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MIL-P-80186	MIL-P-80186A
Class I	Class I
Class II	Class II
	Class III (New class)
Size 1	Size 1
	Size 1a (New size)
Size 2	Size 2
Size 3	Size 3
Size 4	Size 4
Size 5	Size 5
Size 6	Size 6
Size 7	Size 7

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Review activities:
Army - AR
Air Force - 84

User activities:
Army - SM
Navy - MC, OS

Preparing activity:
DLA-IP

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION

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4. TYPE OF ORGANIZATION *(Mark one)*

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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